Introduction

1. Let $f(x) = 2x$. Consider the following function:

☐Regrade

$$F(x) = \int_0^x 2t\ dt.$$

(a) Draw a graph of $f(x)$. On it, illustrate $F(1)$, $F(2)$ and $F(3)$.

(b) What does $F(x)$ measure?

(c) Without further computation, how do you know $F(x)$ is increasing? Concave up?

(d) Calculate $F(1)$, $F(2)$, $F(3)$ and $F(-1)$.

(e) What function do you think $F(x)$ is?

(f) Use the FTC to prove your hypothesis from the previous question, then fill in the blanks below:

$$\frac{d}{dx}\int_0^x 2t\ dt = _____, \text{ so } \int_0^x 2t\ dt \text{ is an } \underline{\text{antiderivative}} \text{ of } f(x) = 2x.$$

(g) Is $G(x) = \int_2^x 2t\ dx$ also an antiderivative of $f(x) = 2x$? If so, what constant do $F(x)$ and $G(x)$ differ by?

Stuff from the Past that will be Important Today!

2. □Regrade

⋆ **Extreme Value Theorem (105L Worksheet 11-3):** If $f(t)$ is continuous on the closed interval $[a, b]$, then it has a _____ and a _____ on that interval.

⋆⋆ **Bounding Integrals (106L Worksheet 6-3):**
If $m \leq f(t) \leq M$ for $a \leq t \leq b$, then

$$_____ \leq \int_a^b f(t)dt \leq _____$$

FTC II

The Second Fundamental Theorem of Calculus
Let f be continuous on an interval. Then for x and a in that interval

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

3. □Regrade **Proof:** Suppose $f(t)$ is a continuous function and let $g(x) = \int_a^x f(t)dt$.

(a) Then

$$\frac{g(x+h)-g(x)}{h} = _____ = _____$$

(b) By ($\star$), since $f(t)$ is a continuous on $[x, x+h]$, then it attains a smallest value m and a largest value M. Then, by ($\star\star$),

$$____ \leq \int_x^{x+h} f(t)dt \leq ____$$

(c) Dividing everything by h, we get

$$____ \leq \frac{\int_x^{x+h} f(t)dt}{h} \leq ____$$

(d) As $h \to 0$, what happens to the interval $[x, x+h]$?

(e) As $h \to 0$, what happens to m and M?

(f) Therefore,

$$\frac{d}{dx}g(x) = \lim_{h\to 0}\frac{\int_x^{x+h} f(t)\,dt}{h} = _____.$$

4. Recall that we were never able to antidifferentiate e^{-x^2}. Can you now write down an antiderivative for it?
□Regrade

5. Find the following derivative in two different ways: $\dfrac{d}{dx}\displaystyle\int_2^x \cos t\, dt$
□Regrade

(a) Using FTC I:

(b) Using FTC II:

6. Let $g(x) = \displaystyle\int_1^x \sqrt{1+t^2}\, dt$.
□Regrade

(a) What is $g'(x)$?

(b) What is $g(x^3)$?

(c) What is $\frac{d}{dx}g(x^3)$? (Hint: you need the chain rule here. This is a composite function.)

7. (a) Find a function $g(x)$ such that $g'(x) = \sqrt{1+x^2}$ and $g(2) = 0$.

☐Regrade

(b) Find a function $g(x)$ such that $g'(x) = \sqrt{1+x^2}$ and $g(2) = 10$.

8. Let $g(x) = \int_0^x f(t)\,dt$, with $f(x)$ is continuous. Crost out the wrong answer for each ☐Regrade of the following:

- If $f(x) > 0$ and $x > 0$, then $g(x)$ is positive/negative and increasing/decreasing.
- If $f(x) > 0$ and $x < 0$, then $g(x)$ is positive/negative and increasing/decreasing.
- If $f(x) < 0$ and $x > 0$, then $g(x)$ is positive/negative and increasing/decreasing.
- If $f(x) < 0$ and $x < 0$, then $g(x)$ is positive/negative and increasing/decreasing.

9. What constant do the following two antiderivatives of $f(x)$ differ by?

☐Regrade

$$F(x) = \int_{-1}^{x} f(t)\,dt \qquad G(x) = \int_{1}^{x} f(t)\,dt$$

(Hint: draw pictures, and see Worksheet on Properties of Integrals, property 4!)